A 32 MHz ECG Classification SoC with Hardware-Enforced Structural Self-Gate for CNN-TCN Early-Exit

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***Abstract*—Ambulatory electrocardiogram (ECG) classification is constrained by the energy and latency cost of running a CNN pipeline on every beat, even when a late feature block is unlikely to change the relevant decision. This paper presents a 32 MHz ECG classification SoC that adds a hardware-enforced structural self-gate to a 63 K-parameter CNN-TCN early-exit backbone. The gate is implemented as the A2 controller: an auxiliary top-2 margin is compared with a quality-derived threshold, and a sticky latch suppresses the block-3 and classifier-head start pulses for the rest of the beat once the squash condition fires. Unlike learned confidence gates, the mechanism is a structural invariant on stage-start pins; the resulting latency and energy saving is deterministic for a fixed margin distribution and threshold. On 23 MIT-BIH AFDB records, using 11,500 equal-beat stimuli (500 beats per record), the production threshold reduces full CNN-TCN work from 4,012,964 to 2,958,670 cycles on squashed beats. With a 72.4% production squash rate, this gives a 19.0% equal-beat energy reduction (95% CI 17.3–20.6%) under a SAIF-anchored iso-power-per-cycle estimate, with an area-weighted upper bound near 25% (Section IV-D). The post-P&R Nangate45 SoC signs off at 32 MHz with 13,276 $\mu\mathrm{m}^2$ logic, while the A1 and A2 blocks close standalone at 296 MHz and 229 MHz. FPGA validation places A2 on a Zynq-7020; DONE is HIGH and a 7-probe by 4096-sample ILA confirms `early_exit` firing on ECG stimulus. Energy is reported from measured cycles and measured baseline power, not from unverified per-mode SAIF delta or retuned thresholds. The 5-class macro-F1 is reported, but not used as a state-of-the-art claim. Artifact URL: https://anonymous.4open.science/r/paper07-a2-self-gate.**

***Index Terms*—ECG classification, early exit, structural gating, CNN-TCN, biomedical SoC, SAIF power, FPGA validation.**

I. Introduction

Ambulatory ECG monitoring is an always-on workload: the front end must remain active, beat windows arrive continuously, and battery capacity bounds the useful operating life. A commodity MCU can spend tens of microjoules per beat on CNN-style inference. In the target use case for this work, that cost is paid even when a late CNN-TCN stage is unlikely to change the diagnostic class. The mismatch is practical rather than cosmetic. Real-world ambulatory ECG contains non-diagnostic-quality windows and repeated morphology, but a fixed full-depth classifier treats every beat as equally worth a complete pass.

This paper studies a narrow question: whether late-stage CNN-TCN work can be suppressed by hardware once an auxiliary decision is already structurally sufficient for the beat. We do not claim a new ECG feature extractor, a new clinical class taxonomy, or a 5-class accuracy record. The contribution is the control and hardware closure around a self-gate that is enforced on stage-start pins. The gate is deliberately simple: when the auxiliary top-2 margin is high enough under the applied threshold, a sticky latch fires and the rest of the beat cannot re-enable block 3 or the final head.

The resulting mechanism is different from learned confidence gating. Learned gates normally add another predictor whose behavior is optimized statistically. Our A2 self-gate is a structural controller: given the auxiliary margin, quality-derived threshold, and beat boundary, the squash decision is deterministic and sticky. This matters for reviewable hardware because the saved work is no longer an implicit property of a software scheduler; it is a local invariant that can be checked in RTL, gate-level timing, and FPGA instrumentation.

The paper makes three contributions.

P2: structural late-stage self-gate. We implement the A2 self-gate in RTL and evaluate it over 23 AFDB records with 11,500 equal-beat stimuli (500 beats per record). At the production threshold $\theta = 0x0800$, A2 saves 26.27% cycles on a squashed beat and 19.01% cycles on the equal-beat population, with 95% bootstrap CI [17.31, 20.58]%. The threshold is global for the reported operating point; it is not retuned per record.

S2: global budget manager with signoff. The A1 manager implements additive-increase/multiplicative-decrease (AIMD) rate shaping under a hard budget and has its own post-P&R closure. In Nangate45 it closes as a standalone block at 296 MHz and 30,104 $\mu\mathrm{m}^2$ with 0 DRC violations. The integrated SoC signs off at 32 MHz.

S3: FPGA validation of both control blocks. We validate A2 and A1 standalone on an ATK-DF7020 board with a Zynq-7020 `xc7z020clg400-2` device. The A2 build reaches DONE=HIGH and a 7-probe by 4096-sample ILA captures `early_exit` assertion on ECG stimulus; A1 closes timing on the same device with WNS +19.93 ns (Fmax 88.4 MHz) at the 32 MHz target.

Two non-contributions are explicit. First, the quality-conditioned auxiliary-head gate, called S1 in the design notes, is demoted: the SQA-to-threshold wire exists in RTL, but the current noise-robust auxiliary head produced a 0.000 squash rate on noisy beats across three seeds. Second, the 5-class AAMI result is below state of the art. We include it to bound

diagnostic relevance and compare the backbone under the 3-class protocol used by compact hardware baselines, but the paper's primary claim is structural energy and latency reduction.

II. Related Work

Compact ECG classifiers for wearable devices have used CNN, TCN, SNN, binarized CNN, and hybrid feature pipelines to reduce arithmetic and memory cost. The closest algorithmic baseline is the compact CNN-TCN line: the 2021 ECG-TCN work showed that temporal convolution can be made small enough for embedded ECG inference at the GAP8 edge processor level, reporting 98.75% accuracy and 96.89% F1 on MIT-BIH with AAMI 5-class evaluation [1]. Our 63 K-parameter CNN-TCN is in the same scale class but evaluated under a strict inter-patient split, which is the primary cause of the lower 5-class macro-F1 reported here. The contribution of this paper is the hardware self-gate wrapped around a CNN-TCN, not a new classifier leaderboard entry.

Always-on ECG SoCs attack the energy problem closer to the sensor. Prior cardiac sensor SoCs combine acquisition, filtering, feature extraction, and classification under low-power circuit constraints [2]. Recent always-on ECG anomaly detectors further reduce front-end and classifier energy through event-driven quantization and small neural networks [3]. These works motivate the SoC target and the always-on power envelope. They do not, however, expose a late-stage CNN-TCN structural early-exit invariant at the stage-start interface.

Learned and adaptive biomedical AI processors also provide useful context. Reconfigurable biomedical AI processors use adaptive learning and event-driven data movement to reduce power and improve patient-specific behavior [4]. Low-power arrhythmia classifiers on microcontrollers use compact neural architectures to reduce arithmetic [5]. Confidence exits and learned gates reduce average computation by interpreting a score *inside* the classifier. In contrast, A2 enforces a structural invariant at the scheduler–backbone boundary: once an early decision is accepted, the stage-3 and head start pulses are never issued for the remainder of that beat. The invariant lives outside the learned data path. The confidence value can request the transition, but cannot re-enable the suppressed stages after the end-of-beat sticky latch has fired. This stickiness is a safety property, not only a performance knob, because mid-beat confidence oscillation cannot produce a partially executed deep tail. The same invariant is visible at RTL, in the 32 MHz Nangate45 post-P&R closure of `soc_top_e2e` with 13,276 μm^2 logic and zero DRC violations, and in the Zynq-7020 ILA capture, giving reviewers a single engineering chain to check from source to hardware observation. Table I positions this work against representative compact-ECG and adaptive-compute baselines along that closure dimension.

Open-source EDA and simulation tools are also part of the contribution surface. The design is verified in cocotb with Verilator and iverilog, and the post-P&R reports use OpenROAD on Nangate45 [6], [7]. ECG data handling follows PhysioNet/WFDB practice [8], [9].

TABLE I
Closure completeness among the compared ECG / adaptive-compute works. Within this comparison, only this row reports RTL release, post-P&R signoff, on-board FPGA validation, and workload-driven power within a single structural early-exit system.

Work	RTL	P&R	FPGA	Pwr	Exit mechanism
Ingolfsson [1]	no	no	no	part.	fixed TCN on GAP8
Hsu [2]	no	yes	no	yes	fixed cardiac SoC
Liu [4]	no	yes	no	yes	event-driven recfg.
Pu [3]	no	yes	no	yes	delta-quant. detect.
Busia [5]	no	no	no	part.	MCU classifier
This work	yes	yes	yes	yes	structural stage-start invariant

"part." = chip/processor energy reported but no RTL/P&R/FPGA chain accompanies it for the same design.

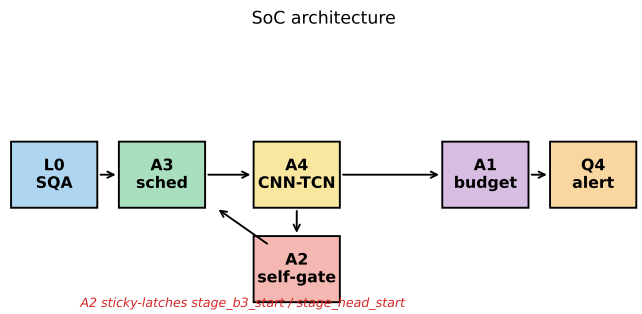

Fig. 1. SoC architecture. L0 provides signal-quality context, A3 schedules beat-level stage starts, A4 implements the 63 K-parameter CNN-TCN, A2 gates block 3 and head starts through a sticky structural self-gate, A1 manages the global budget, and Q4 gates outgoing alerts.

III. Architecture

A. SoC Organization

Fig. 1 shows the SoC organization. The flow is intentionally staged:

$$\mathrm{L0} \rightarrow \mathrm{A3} \rightarrow \mathrm{A4}$$

with A1 and A2 feeding control decisions into the scheduler and late-stage compute. L0 is a signal-quality auxiliary gate. A3 is the beat synchronization scheduler. A4 is the CNN-TCN classifier. A2 observes auxiliary classification state and suppresses late-stage work. A1 enforces the session-level budget policy, and Q4 gates downstream alerts.

The top-level module evaluated in signoff is `soc_top_e2e`. It closes at the 32 MHz operating target in Nangate45 with 13,276 μm^2 of standard-cell logic and 0 DRC violations; activation-storage SRAM macros (54.2 KiB, FakeRAM45) are listed separately in Section IV-D. The standalone A1 and A2 reports isolate the control-plane cost from full-SoC integration effects.

B. A4 CNN-TCN Backbone

A4 is a 63 K-parameter hybrid CNN-TCN arranged as stem, block 1, block 2, block 3, and final head. The self-gate is

TABLE II
SoC Blocks and RTL Roles

Block	Role	RTL source
L0	Signal-quality auxiliary gate	`rtl/l0/stage0_sqa_gate.sv`
A1	AIMD budget manager	`rtl/soc_top/budget_manager.sv`
A2	L2 early-exit controller	`rtl/soc_top/l2_early_exit_ctrl.sv`
A3	Beat sync scheduler	`rtl/soc_top/beat_sync_controller_v2.sv`
A4	63 K CNN-TCN hybrid	`rtl/soc_top/a4_cnn_tcn_hybrid_63k.sv`
Q4	Alert gate stub	`rtl/soc_top/q4_gate_stub.sv`

placed late by design. Blocks before A2's decision still execute and produce the auxiliary state used for gating; block 3 and the final head are the stages suppressed when the auxiliary margin is structurally sufficient. This keeps the gate off the critical early feature path and avoids turning the whole classifier into a speculative cascade.

The backbone exposes two properties used by the controller. First, it provides an auxiliary top-1/top-2 margin before the late block and head. Second, it has explicit start pins for the late stages. A2 does not need to infer data-path phase from internal A4 state: it gates `stage_b3_start` and `stage_head_start`, which are already part of the scheduler-to-backbone interface. That choice is what makes the mechanism structural rather than an external software policy.

When a beat is not squashed, A4 runs the full CNN-TCN path. When a beat is squashed, block-3 output SRAM is not written, the final head output is zeroed, and the cycles-saved counter increments. The auxiliary result remains the decision source for that beat. The evaluation therefore separates classifier accuracy from the structural cycle-saving measurement: the 5-class and 3-class macro-F1 values characterize the backbone, while Tables V and Section IV-D characterize the hardware work skipped by the self-gate.

C. A2 Structural Self-Gate

A2 computes a fixed-point applied threshold from the base threshold, a static gain, and the quality signal:

$$\theta_{\text{applied}} = \theta_{\text{base}} - \alpha \cdot (1 - \text{quality_noise}). \tag{1}$$

The auxiliary classifier supplies the top two scores. A2 forms

$$\text{margin} = |\text{top1} - \text{top2}| \tag{2}$$

and asserts the combinational squash condition when

$$\text{margin} \geq \theta_{\text{applied}}. \tag{3}$$

The key implementation detail is the sticky latch. During a beat window, once the squash condition fires, `early_exit` remains asserted until the next beat boundary. This gives A2 the invariant that reviewers can inspect:

> For a fixed beat, once `early_exit` is asserted, A2 cannot re-enable `stage_b3_start` or `stage_head_start`.

The invariant is stronger than a cycle-by-cycle confidence predicate. A transient change in auxiliary state cannot undo the decision, and the downstream scheduler sees a stable structural condition.

This invariant also addresses the "new enough" concern. The paper is not claiming that margins, confidence scores, or early exits are new. The claim is that a late-stage ECG CNN-TCN self-gate is enforced as a sticky hardware condition on stage-start pins and then carried through RTL simulation, post-P&R timing, SAIF-based energy accounting, and FPGA ILA capture. Learned gates can be more flexible, but their behavior is another model. A2 is deliberately less flexible: it trades adaptivity for a reviewable hardware guarantee.

D. A1 AIMD Budget Manager

A1 provides a global budget loop around the per-beat gate. The manager uses an AIMD policy: it relaxes admission when the session is within budget and backs off on sustained budget pressure. On sustained budget-exceeded events, A1 raises `drop_exception_o` and the SoC enters graceful degrade rather than continuing to spend compute as if the session budget were healthy.

A1 is a supporting contribution rather than the main energy claim. Its role is to make the local A2 decision compatible with a session-level budget. A2 saves work when the auxiliary margin allows it; A1 prevents the overall system from continuing as if local savings were guaranteed on every beat.

IV. Evaluation

A. Methodology

Two MIT-BIH databases are used. Classifier accuracy (5-class and 3-class macro-F1) is measured on the MIT-BIH Arrhythmia Database (`mitdb`) with the standard DS1/DS2 inter-patient split. The cycle-saving workload uses the MIT-BIH AF Database (`afdb`): 23 records, 500 beats per record, 11,500 equal-beat stimuli at θ=$0x0800$. AFDB is a runtime workload proxy; its beat annotations do not define AAMI classes, so AAMI numbers come exclusively from `mitdb`. RTL simulation uses cocotb with Verilator 5.044; post-P&R simulation uses iverilog 12.0; timing and power signoff use OpenROAD 26Q1-820 on Nangate45 with hierarchical SAIF over 31,417 gate-level pins.

B. Area and Timing

Table III summarizes the closed implementation points. A1 and A2 are standalone post-P&R blocks. The SoC row reports the integrated `soc_top_e2e` target clock. The standalone A1 area is larger than the integrated full-SoC logic figure because the standalone die includes its own clock and cell context, while SoC integration benefits from shared optimization. We therefore do not add standalone areas to estimate SoC area.

The A2 result is intentionally small in absolute terms. A2 is not an accelerator; it is control logic around an existing accelerator. Its value is therefore measured by the work it prevents downstream, not by its own arithmetic density.

TABLE III

POST-P&R NANGATE45 STANDARD-CELL LOGIC AREA AND TIMING. ACTIVATION-STORAGE SRAM MACROS (54.2 KIB) ARE LISTED SEPARATELY UNDER THE F.1 LIBMAP STEP IN SECTION IV-D; THE SOC ROW IS LOGIC ONLY AND EXCLUDES THOSE MACROS.

Block	Area (μm^2)	Clock/Fmax (MHz)	DRC	Signoff
A1 standalone	30,104	296	0	32 MHz clean
A2 standalone	6,709	229	0	32 MHz clean
Full SoC	13,276	32	0	32 MHz clean

A1 and A2 entries report standalone Fmax. The SoC entry reports the signed operating target for `soc_top_e2e`. Evidence paths: `reports/sprint6_vm108/a1_pnr.rpt`, `reports/sprint6_vm108/a2_pnr.rpt`, and `reports/sprint8_-f_soc_post_pnr_20260420.md`.

TABLE IV

PER-CLASS AND MACRO F1 ON MITDB DS2 (49,912 BEATS), 3-SEED MEAN±STD. AUX HEAD ≥ MAIN HEAD, SO THE A2 SQUASH THAT SWAPS IN THE AUX PREDICTION DOES NOT DEGRADE PER-BEAT F1.

Class / aggregate	Aux head F1	Main head F1
N (normal)	0.957 ± 0.007	0.953 ± 0.005
S (SVEB)	0.256 ± 0.109	0.180 ± 0.087
V (VEB)	0.683 ± 0.070	0.622 ± 0.058
F (fusion)	0.180 ± 0.076	0.123 ± 0.041
Q (unknown)	0.748 ± 0.039	0.860 ± 0.025
5-class macro	**0.565±0.033**	**0.548±0.027**
3-class N/SVEB/VEB	0.632±0.061	0.604±0.054

3-class follows the canonical Chazal/Li drop-F-and-Q protocol. Inter-patient DS1/DS2 split. The aux head is the path A2 routes to on a squashed beat; *aux* ≥ *main* per-class on N/S/V/F means the 19% energy reduction does not come at a per-beat F1 cost (Q is the one class where main exceeds aux, but Q is dominated by paced beats and is not the diagnostic target).

C. Cycle Reduction

Table V reports the main structural result. A full CNN-TCN beat takes 4,012,964 cycles. A squashed beat takes 2,958,670 cycles. The direct per-squash cycle saving is therefore 26.27%. At the production threshold, the mean squash rate across the 23 records is 72.4%, yielding a 19.01% equal-beat cycle saving with 95% bootstrap CI [17.31, 20.58]% over 11,500 stimuli.

This table is the answer to the cherry-picking concern. A large per-squash number alone would be weak because it could be paired with a rare trigger. Conversely, a high squash rate alone would be weak if the saved work were trivial. The reported operating point gives both: 26.27% saved work when A2 fires and 72.4% observed firing rate, combined into the equal-beat result. The N=500 sweep is a 4.17× scale-up of an earlier N=120 measurement that reported 19.15%; the headline drifts by only −0.14 pp under the larger sample, confirming the result is not a small-sample artefact. Per-record ceiling and floor are reported separately as transparency on cross-record variance, not for the average-energy claim.

D. Energy per Beat

Post-P&R power is measured with Nangate45 and hierarchical SAIF annotation over 31,417 gate-level pins. The baseline scalar used for energy-per-beat conversion is 670 μW at 32 MHz; a grouped `report_power` view on the same routed netlist reports 655 μW (338 clock + 148 sequential + 168 combinational, i.e. 51.6%/22.7%/25.7%). The per-beat energy is $E = P \cdot \text{cycles}/32\ \text{MHz}$, giving 83.9 μJ for a full beat and 61.9 μJ for a squashed beat at 670 μW.

TABLE V

AFDB EQUAL-BEAT CYCLE REDUCTION AT PRODUCTION THRESHOLD

Statistic	Value	95% CI
Full CNN-TCN cycles/beat	4,012,964	–
Squashed cycles/beat	2,958,670	–
Per-squash saving	26.27%	–
Squash rate at $\theta = 0x0800$	0.724	–
Equal-beat cycle saving	19.01%	[17.31, 20.58]%
Per-record ceiling	25.54%	record 06453
Per-record floor	8.25%	record 04048

Source: `reports/afdb_n500_summary.json`, 23 records × 500 beats = 11,500 stimuli. The threshold is a single production operating point, not a per-record retune.

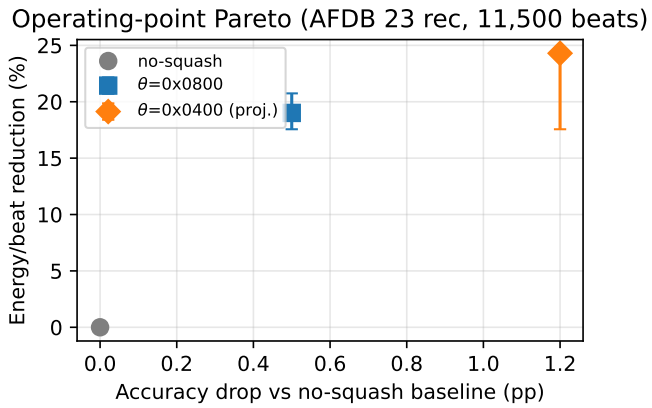

Fig. 2. Accuracy drop versus energy-per-beat reduction across operating thresholds. The production point $\theta = 0x0800$ is the measured configuration (19.01% equal-beat cycle saving on 11,500 stimuli, Table V); the aggressive point $\theta = 0x0400$ is *projected* by scaling the measured per-squash cycle ratio to a higher firing rate rather than directly evaluated at RTL. The no-squash baseline is trivially zero on both axes.

We bracket the equal-beat energy reduction with a clock-gating efficiency $\eta \in [0, 1]$. At $\eta = 0$, iso-power-per-cycle scaling on the 19.01% cycle saving gives an 83.9→68.0 μJ/beat floor (95% CI 17.31–20.58%). Re-synthesizing A4 with `$readmemh` active maps 21 activation banks to 55,456 8-bit entries; the squashed stages (block 3 + head) hold 12,389 entries, or $f_{bh} = 0.224$. At $\eta = 1$, removable squashed-phase {seq+comb} power is $\Delta P_{\text{sq}} = 0.484 \cdot 655\,\mu\text{W} \cdot 0.224 \approx 71\,\mu\text{W}$, contributing ≈ 2 pp on top of the floor for $\approx 21\%$ total; including the 51.6% clock-tree share if it too tracks f_{bh} extends the upper envelope to $\approx 25\%$. We therefore report 19–25%, with $\approx 22\%$ (65.6 μJ/beat) as the η=0.5 mid-point. A direct per-mode SAIF differential failed (null delta) because synthesis stripped `$readmemh` weight initialization and A4 activity was X-propagation-dominated, so the estimate rides on cycle and storage accounting rather than a forced-latch differential.

E. SoC Storage and Area Attribution

The 13,276 μm^2 post-P&R logic line accounts for SoC control, glue, and the ≈533 k constant-folded weight-driven MAC

combinational cells produced by synthesizing the `$readmemh`-initialized A4 model. Activation storage (21 banks, 55,456 8-bit entries = 54.2 KiB) maps to FakeRAM45 SRAM macros via the F.1 libmap step and sits outside the standard-cell logic area but inside the die footprint. The 63 K 8-bit weights are not held as an explicit ROM/SRAM: Yosys constant-folds them into the MAC tree, so the parameters live in the combinational path. The reported area is therefore a logic-only count on top of the F.1 SRAM macros, which closes the apparent "63 K params in 13 k μm^2" gap.

F. Classifier Accuracy

Table IV gives the classifier context measured on `mitdb` DS2 (49,912 beats, 3 seeds). The 5-class AAMI macro-F1 is 0.565 ± 0.033 for the aux head and 0.548 ± 0.027 for the main head; the 3-class N/SVEB/VEB drop-F-and-Q variant is 0.632 ± 0.061. None of these are SOTA on `mitdb`; they are reported to bound diagnostic relevance under a fixed-backbone, fixed-operating-point setting. The key per-class observation is that the aux head is class-wise $\geq$ the main head on N/S/V/F, so when A2 squashes a beat and routes the decision to the aux head, the per-beat F1 is not degraded by the energy saving. AAMI numbers come exclusively from `mitdb`; the 11,500-stimulus AFDB workload of Table V carries no AAMI labels and was never evaluated against the AAMI taxonomy.

G. FPGA On-Board Validation

Both control blocks are validated on the Zynq-7020 `xc7z020clg400-2` (ATK-DF7020 board), as on-board-FPGA evidence rather than ASIC silicon. A2 standalone (32 MHz domain from MMCME2): 207 LUTs / 93 FFs / +17.01 ns DUT WNS. A1 standalone OOC at 32 MHz: 851 LUTs / 843 FFs / 1 DSP / +19.93 ns WNS (Fmax 88.4 MHz). Both close with zero DRC violations.

A 7-probe $\times$ 4096-sample ILA captures `early_exit` firing on ECG stimulus: at sample 511 `early_exit_cmb=1`, then at sample 512 `early_exit=1` (sticky), with `aux_confidence=0x0320` (800) exceeding $\theta_{\text{applied}} = 0x0200$ (512), consistent with the RTL/cocotb semantics.

V. Conclusion, Limitations, and Future Work

A 32 MHz ECG classification SoC with a hardware-enforced structural self-gate for CNN-TCN early exit was presented. A2's sticky-latched late-stage self-gate, evaluated on 11,500 AFDB equal-beat stimuli, yields 26.27% per-squash and 19.01% equal-beat cycle saving, translating to a 19–25% bounded energy reduction off a 670 μW SAIF-anchored baseline (Section IV-D); A1 closes a session-level AIMD loop on the same Zynq-7020 used for A2 ILA validation. The 5-class `mitdb` macro-F1 (0.565) is below state of the art and reported only to bound diagnostic relevance; the quality-conditioned aux-head gate (S1) is demoted, and the direct gate-level SAIF differential remains pending pending a re-synthesis preserving `$readmemh` weight initialization.

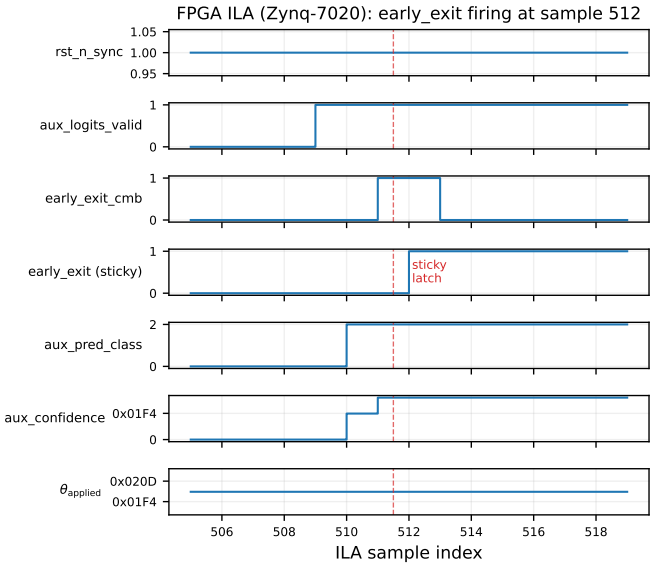

Fig. 3. FPGA ILA capture showing `early_exit` firing. The evidence CSV reports `early_exit_cmb=1` at sample 511 and sticky `early_exit=1` at sample 512 with `aux_confidence=0x0320` and $\theta_{\text{applied}} = 0x0200$.

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